Definition 7.9.4. An abelian category $\mathcal{C}$ is **semisimple** if any short exact sequence in this category splits, i.e., is isomorphic to a sequence

$$0 \to X \to X \oplus Y \to Y \to 0$$

(where the maps are obvious).

Example 7.9.5. The category of representations of a finite group G over a field of characteristic not dividing $|G|$ (or 0) is semisimple.

Note that in a semisimple category, any additive functor is automatically exact on both sides.

Example 7.9.6. (i) The functors Ind_K^G, Res_K^G are exact.

(ii) The functor $\mathrm{Hom}(X, ?)$ is left exact, but not necessarily right exact. To see that it need not be right exact, it suffices to consider the exact sequence

$$0 \to \mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$$

and to apply the functor $\mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, ?)$.

(iii) The functor $X \otimes_A$ for a right A-module X (on the category of left A-modules) is right exact but not necessarily left exact. To see this, it suffices to tensor multiply the above exact sequence by $\mathbb{Z}/2\mathbb{Z}$.

Exercise 7.9.7. Show that if (F, G) is a pair of adjoint additive functors between abelian categories, then F is right exact and G is left exact.

Exercise 7.9.8. (a) Let Q be a quiver and let $i \in Q$ be a source. Let V be a representation of Q and let W be a representation of $\overline{Q_i}$ (the quiver obtained from Q by reversing arrows at the vertex i). Prove that there is a natural isomorphism between $\mathrm{Hom}\left(F_i^- V, W\right)$ and $\mathrm{Hom}\left(V, F_i^+ W\right)$. In other words, the functor F_i^+ is right adjoint to F_i^-.

(b) Deduce that the functor F_i^+ is left exact and F_i^- is right exact.